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20AM4S01: MOBILE APP DEVELOPMENT				

COURSE OBJECTIVES:

- To understand the components and structure of mobile application development frameworks for Android and windows OS based mobiles.
- To understand how to work with various mobile application development frameworks.
- To learn the basic and important design concepts and issues of development of mobile applications.
- To understand the capabilities and limitations of mobile devices.

COURSE OUTCOMES:

At the end of this course, students will be able to:

CO1: Identify various concepts of mobile programming that make it unique from programming for other platforms.

CO2: Critique mobile applications on their design pros and cons.

CO3: Utilize rapid prototyping techniques to design and develop sophisticated mobile interfaces.

CO4: Program mobile applications for the Android operating system that use basic and advanced phone features.

CO5: Deploy applications to the Android marketplace for distribution.

LIST OF EXPERIMENTS

1. Introduction to mobile technologies and devices, Android platform and applications overview
2. Setting Android development environments
3. Writing Android applications, Understanding anatomy of an Android application
4. Develop an application that uses GUI components, Font and Colours
5. Develop an application that uses Layout Managers and event listeners.
6. Write an application that draws basic graphical primitives on the screen.
7. Develop an application that makes use of databases.
8. Develop an application that makes use of Notification Manager
9. Implement an application that uses Multi-threading
10. Develop a native application that uses GPS location information
11. Implement an application that writes data to the SD card.
12. Implement an application that creates an alert upon receiving a message
13. Write a mobile application that makes use of RSS feed
14. Develop a mobile application to send an email.
15. Develop a Mobile application for simple needs (Mini Project)

References:

1. B.M. Harwani, “Android Programming unleashed”, Pearson, 2013.
2. Bill Phillips, Chris Stewart, Brian Hardy, Kristin Marsicano, “Android Programming (Big Nerd Ranch Guide)”, Pearson, 2016
3. Hellman by Erik Hellman, “Android Programming – Pushing the limits”, WILEY, 2013

Web References:

1. The Complete Android N Developer Course –Udemy
   https://www.udemy.com/course/complete-android-n-developer-course/?altsc=428526
2. Android Development Courses on Google developers training
   https://developers.google.com/training/android/
3. Mobile Computing - Video course- NPTEL
   https://nptel.ac.in/courses/106/106/106106147/#
4. Android Tutorial – Tutorial Point
   https://www.tutorialspoint.com/android/index.htm